

Real-Time Impact Point Prediction of Suborbital Rockets Using Kalman Filtering

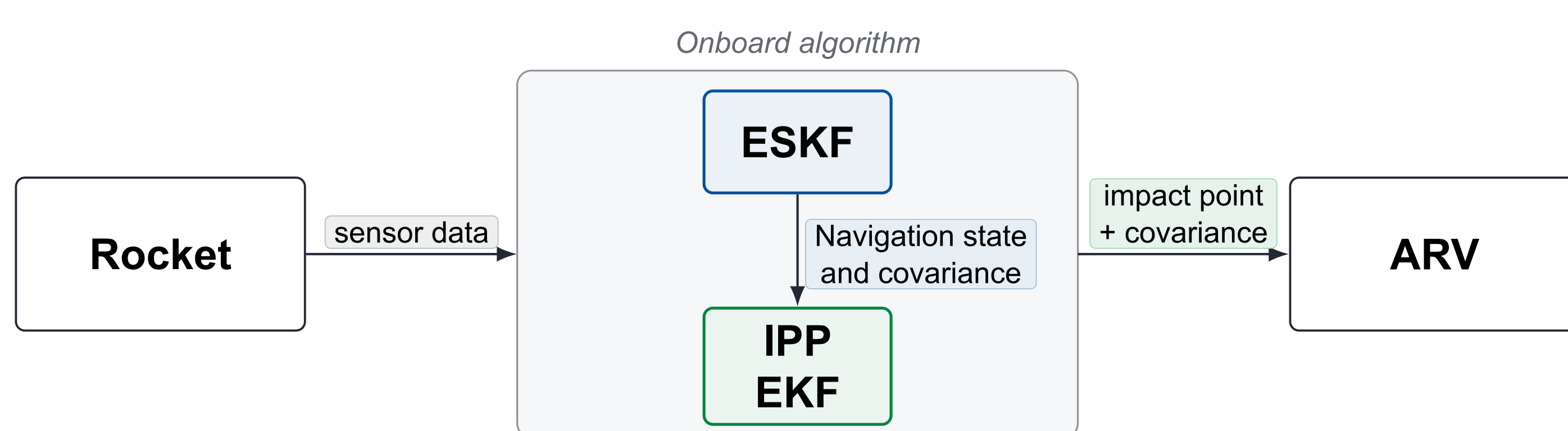
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Research Objective

Rocket launch rates are rising, making sustainable recovery of payloads and first stages increasingly important. Autonomous recovery vehicles (ARVs) address this by reducing cost, human exposure, and environmental footprint. This requires solving two coupled problems. The rocket's landing location must be predicted in real time so the ARV can be guided to that location.

Impact point prediction (IPP) is the real-time estimate of the rocket's landing location and uncertainty, updated continuously from onboard sensor data. An ARV steers to this estimate.



IPP Architecture

Navigation — Error-State Kalman Filter (ESKF)

The ESKF maintains a continuous estimate of position, velocity, attitude, and sensor biases. IMU measurements propagate the state at high rate. GNSS, barometer, and magnetometer measurements then correct it via the error-state injection scheme [3]. The filter outputs a full state vector *and covariance* that are passed directly as the initial condition for the IPP filter.

Impact Point Prediction — Cartesian EKF

The IPP filter propagates a Cartesian state $\mathbf{x} = [\mathbf{p}^T, \mathbf{v}^T]^T$ in NED coordinates forward until altitude reaches sea level:

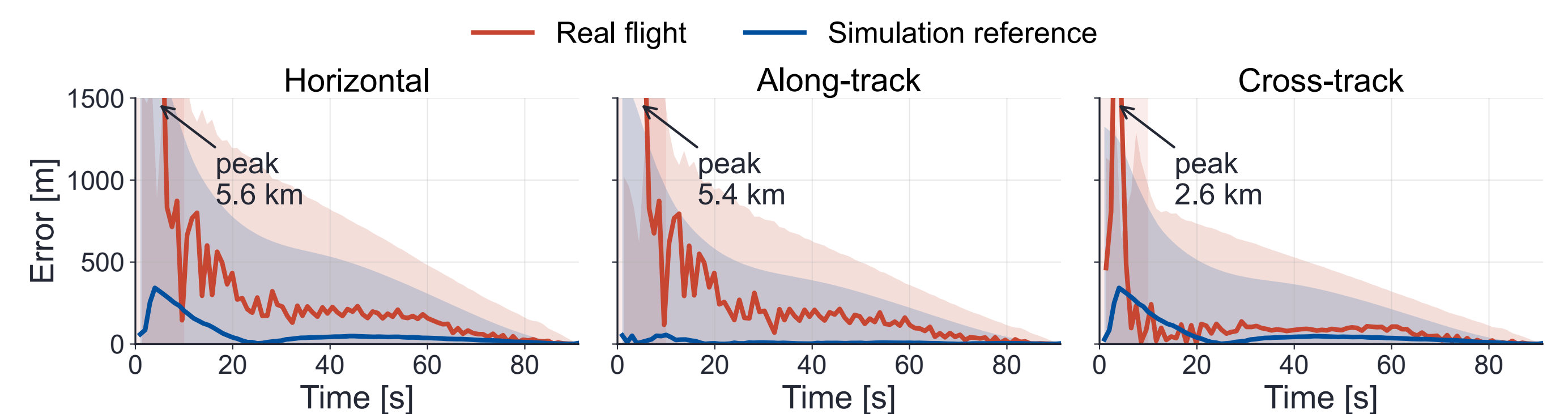
$$\dot{\mathbf{p}} = \mathbf{v}, \quad \dot{\mathbf{v}} = \mathbf{g} - K_D \mathbf{v}_a + \mathbf{R}(\hat{\mathbf{q}}) \mathbf{T}_b, \quad K_D = \frac{\rho C_D A \|\mathbf{v}_a\|}{2m}$$

$\mathbf{v}_a$ airspeed vector, ρ air density, C_D drag coefficient, A reference area, m mass. $\mathbf{R}(\hat{\mathbf{q}})$ rotates the body-frame thrust $\mathbf{T}_b$ to NED using the ESKF attitude quaternion $\hat{\mathbf{q}}$.

The equations are propagated forward until the predicted altitude reaches sea level, yielding a predicted impact point and propagated uncertainty ellipse updated in real time.

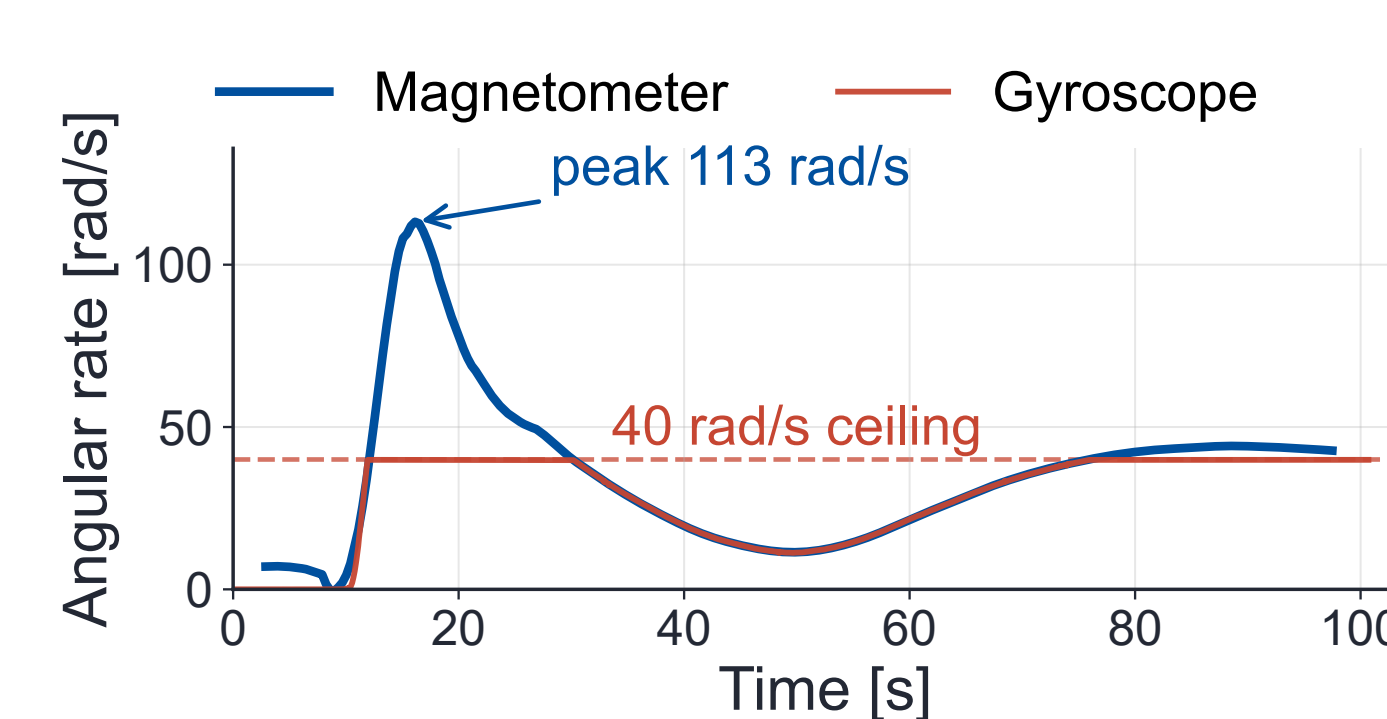
Component	Key design choice
ESKF	Fuses GNSS, barometer, and magnetometer
IPP	Forward propagation of position and velocity using EKF equations
Interface	IPP re-initialised from ESKF state and covariance at each GNSS update

IPP Prediction Error



During the burn phase ($t \lesssim 10$ s) the prediction is corrupted by gyroscope and accelerometer saturation under high spin, vibration, GNSS error, and thrust/mass mismodelling. After burnout the error drops sharply and decays through to impact. Errors are reported against the GNSS reference.

Challenges & Outlook



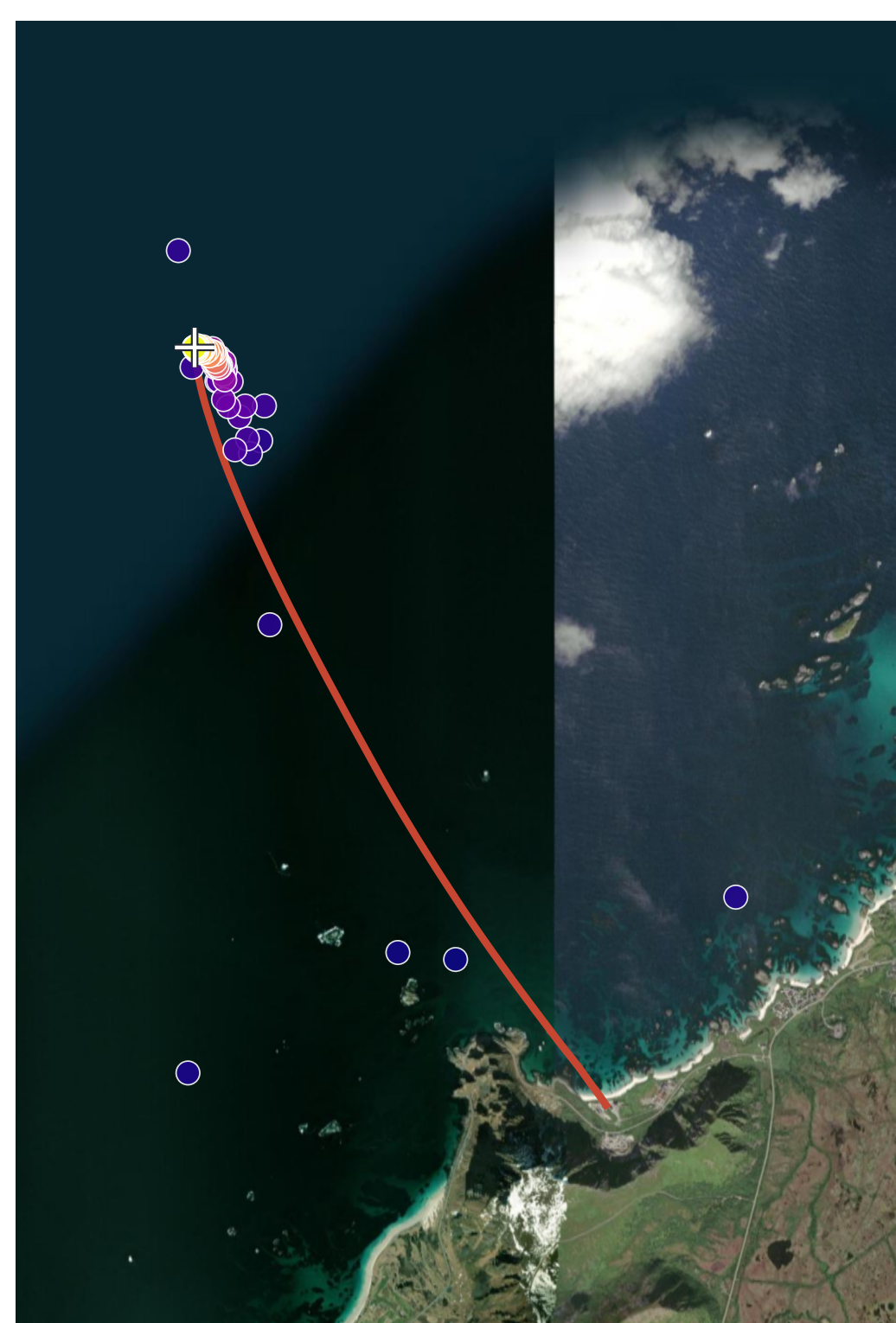
Spin saturation. The gyroscope saturates at ± 40 rad/s through burn. The magnetometer recovers the true ≈ 113 rad/s peak. Mitigations include a higher-range IMU, IMU placement closer to the spin axis, and a lower-spin rocket.

- **Wind and drag uncertainty.** Real-time wind estimation and in-flight drag identification remain open problems for operational use.
- **Known mass and thrust.** The IPP model treats vehicle mass and thrust as deterministic inputs. Propellant burn-off and thrust variations introduce unmodelled uncertainty that limits pre-burnout accuracy. In-flight parameter estimation would address this.
- **Scaling to orbital recovery.** The ESKF and Cartesian architecture requires only higher-grade inertial sensors and a longer propagation horizon to support autonomous ship pre-positioning before first-stage recovery.

References

- [1] K. Gryte et al. *Recovery of Rocket Payloads and First Stages Using Unmanned Vehicles*. IEEE Aerospace Conference, 2025.
- [2] A. Meta. *Impact Point Prediction of High-Power Rockets and Sounding Rockets*. M.Sc. Thesis, NTNU, 2025.
- [3] E. K. Jensen. *Rocket science: Catch space rockets using unmanned vehicles*. M.Sc. Thesis, NTNU, 2026.

Andøya Test Flight



Andøya student-rocket flight (Mongoose 98), 90 s flight to impact. The map shows the rocket ground track (red) and the IPP predictions coloured by flight time, with purple indicating early predictions and yellow indicating late predictions converging toward impact.

The onboard sensor suite comprised an IMU, GNSS, and magnetometer. Barometer data was not available on this flight. Wind profiles were provided by a weather balloon launched 30–45 min before launch. The rocket impacted approximately 6.3 km from the launch pad.

Outlier predictions during the burn phase stem from initial yaw uncertainty and attitude instability caused by gyroscope saturation under high spin, possibly compounded by thrust model errors. Once ballistic, predictions converge toward the actual impact, marked by the crosshair.